

BIOL 40C

Class 9

Body Fluid Compartments

Water Balance & Reyna's Case

Monday, May 4, 2026 | 10:00–11:50 AM | Room 8403

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Today's Roadmap

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CONCEPT

Block 1	Welcome + Reyna — Step 1	10:00–10:25
Block 2	Body Fluid Compartments + Step 2 (Diagnosis)	10:25–10:55
Block 3	Water Balance + Step 3 (Cellular)	11:05–11:30
Block 4	BP, ACE Inhibitors + Step 4 (Treatment)	11:30–11:50

Bridge from Week 3

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CONCEPT

Last week we built a kidney that filters, reabsorbs, and secretes.

This week we slow down right at the end and ask:

**How much water?
(osmolality)**

ADH + thirst

**How much sodium?
(volume)**

RAAS / aldosterone / ANP

And our patient today shares Santiago's anatomy — but her damage took years, not days.

KC Quiz 4 — Hot Spots

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A&P REVIEW

1

ICF is bigger than ECF.

Two-thirds of body water lives inside cells. Plasma is only ~3 L.

2

Tonicity $\neq$ osmolality.

Osmolality counts particles. Tonicity is what happens to a cell in that solution.

3

Osmoreceptors $\neq$ baroreceptors.

Two parallel sensor systems. Water question vs. volume question.

CHECK

If you drink 1 L of pure water, which compartment changes the most?

A. Plasma

B. Interstitial fluid

C. Intracellular fluid

D. All change equally

Vote on Poll Everywhere

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CHECK



Vote at:

PollEv.com/glagna

Or scan the QR code →

Live results display
once everyone votes.

Q: If you drink 1 L of pure water, which compartment changes the most?

BIOL40C-W4-C9-Q1: If you drink 1 L of pure water, which compartment changes the most?



Plasma

0%

Interstitial fluid

0%

Intracellular fluid

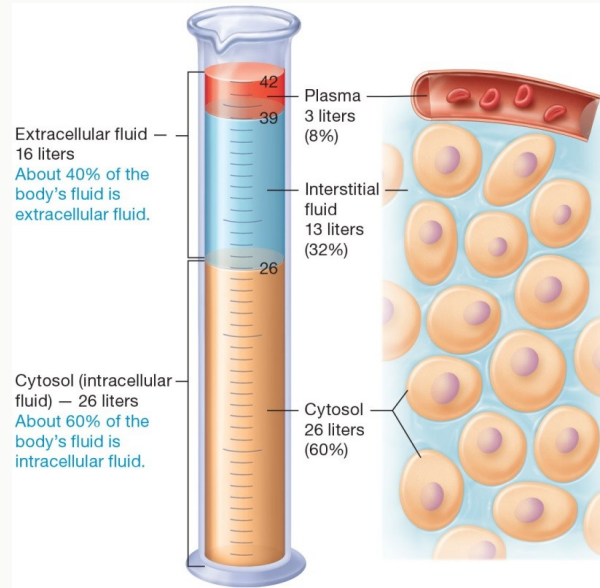
0%

All change equally

0%

Answer: C. Intracellular fluid

Water distributes by mass. ICF is the biggest tank (~26 L of ~42 L total).



Amerman Fig. 25.2 — body fluid compartments by volume

Meet Reyna

Reyna, age 52, comes into your clinic for her yearly physical.

"Mostly, I'm doing well! I've been trying to take care of myself by getting in some exercise and watching my diet, just like you advised me last time. I have to admit, though, that I find myself needing to use the restroom quite a lot these days. Maybe I'm just getting to be an old lady. I also have occasional, mild headaches, but again, I don't think I can complain too much."

On exam: BP elevated. Mild pitting edema in legs.

Think-Pair-Share

ACTIVITY

With your team, take 2 minutes:

- What conditions might Reyna be suffering from?
- What additional information would help narrow the diagnosis?

Things to weigh:

- *Frequent urination + mild headaches + leg edema + high BP: what physiology connects these?*
- *Last week's framework: when does damaged kidney filter cause edema and high BP?*

Open Your Worksheet — AA-2: Reyna's Case

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ACTIVITY

We'll work Reyna's case live in class — across 4 steps, in your team.

Open AA-2 in Canvas → make a copy of the worksheet → log your team's reasoning as we go.

- 1** Open the AA-2 worksheet on Canvas now (laptop or printout)
- 2** Make your own copy (Google Doc or local download)
- 3** Capture your team's reasoning at each of the 4 steps as we work through them today
- 4** Finish the write-up after class. Submit as PDF on Canvas by Sat May 9, 11:59 PM

AA-1 (Pearson IP2 Urine Concentration & Volume Review) is self-paced — do that on your own this week.

Block 2

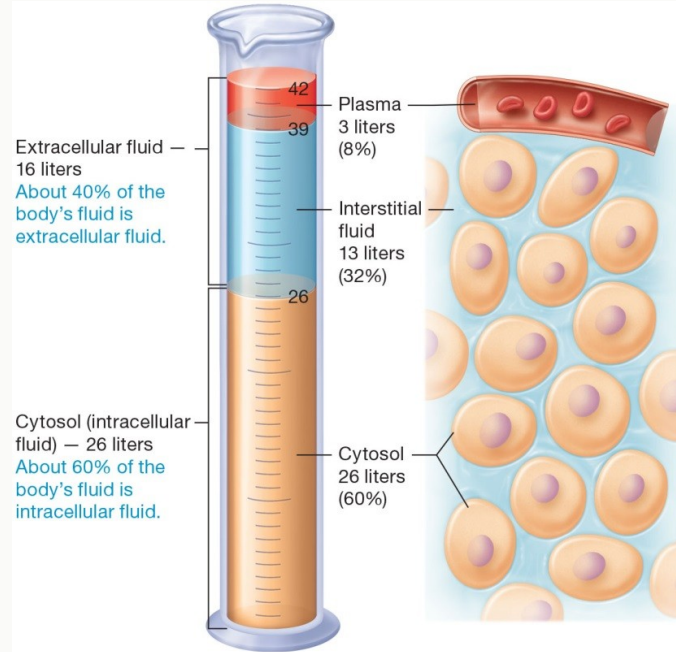
Body Fluid Compartments

Building the framework before we look at the labs

The Three Body Fluid Compartments

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CONCEPT



Amerman Fig. 25.2

Total Body Water (TBW)

≈ 60% body weight (σ^7); ≈ 50% (♀)

ICF (cytosol)

≈ 26 L

high K^+ , low Na^+

Interstitial

≈ 13 L

between cells

Plasma

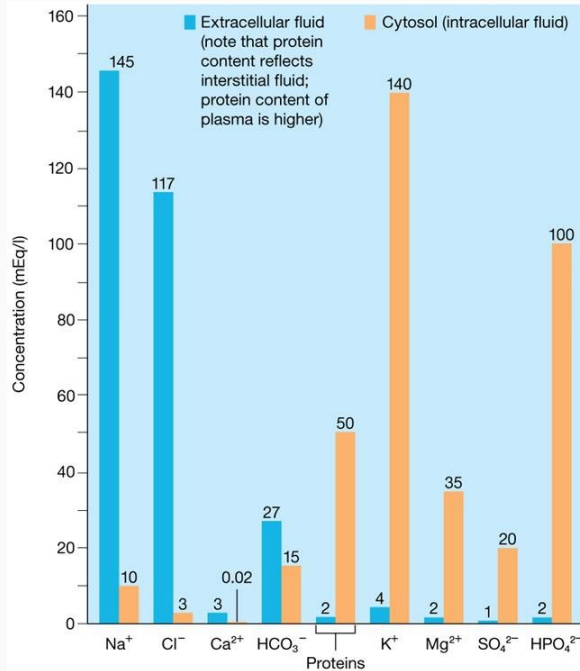
≈ 3 L

in vessels

ECF and ICF Are Chemically Different

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CONCEPT



Amerman Fig. 25.3

Why this matters:

- ECF: Na⁺ dominant
- ICF: K⁺ dominant
- Different barriers do different jobs:

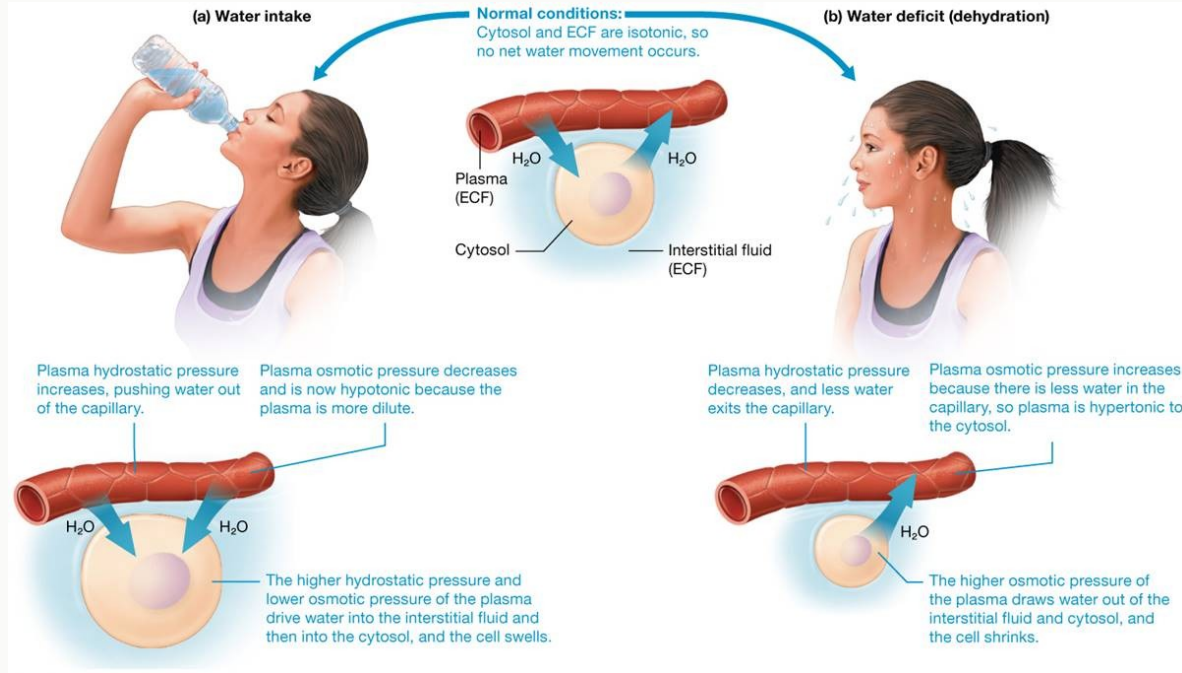
Cell membrane → ICF ↔ ECF
(water free; solutes blocked)

Capillary wall → plasma ↔ interstitial
(water + small solutes free)

Water Movement: Hydrostatic + Osmotic Pressure

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A&P REVIEW



Amerman Fig. 25.4 — water shifts under intake vs. dehydration

Capillary balance:

- Hydrostatic P pushes OUT
- Osmotic π pulls IN

If oncotic/colloidal pressure DROPS...

(albumin lost in urine)

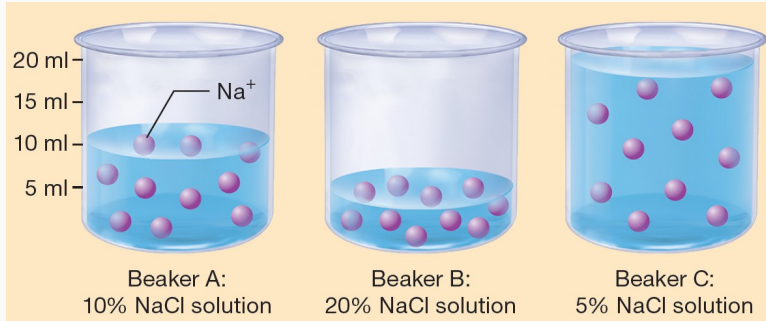
→ Interstitial fluid wins
→ EDEMA

Hold this for Reyna's labs.

Tonicity & Three Common IV Fluids

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CONCEPT



Amerman Fig. 25.5 — three NaCl concentrations

0.9% NaCl (NS)

Isotonic. Stays in ECF.

Volume resuscitation

D5W

Free water (dextrose metabolized).

Free-water deficit

Lactated Ringer's

Isotonic + K⁺, Ca²⁺, lactate.

Trauma, surgery

Note: avoid LR in CKD due to K⁺ load.

Reyna's Test Results

STEP

You also recall that Reyna was diagnosed with **type 2 diabetes** 5 years ago and is on **metformin**.

Test	Reyna's Result
Urinalysis	Albumin: 100 mg / 24 hr (normal < 30)
Blood creatinine / eGFR	eGFR 60 mL/min (normal > 90)
Renal ultrasound	Mildly small, echogenic kidneys
Blood pressure	Elevated

Interpret These Results

ACTIVITY

With your team, work through these (5 minutes):

- 1 What is being measured in the urinalysis and blood tests, and what do those values mean?
- 2 How do urinalysis, blood, and ultrasound results connect to one another?
- 3 Given Reyna's history of diabetes, what condition does this picture suggest?
- 4 Use last week's framework: which structure on the kidney filter is most likely damaged?

Don't ask Giorgio for the answer yet. Reason it out from what you've learned.

Break

10 minutes — back at 11:05

Block 3

Water Balance + Step 3 (Cellular)

Why does Reyna pee so much, and what's happening at the glomerulus?

Two Parallel Sensor Systems

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CONCEPT

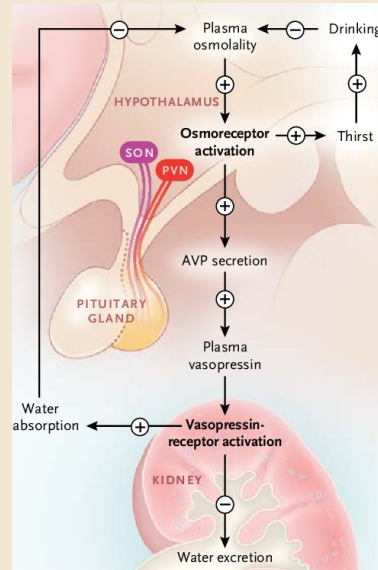
Water question

"Is my plasma osmolality at set point (~285 mOsm/kg)?"

Sensors:
hypothalamic osmoreceptors

Responses:

- ADH (vasopressin)
- Thirst



Volume question

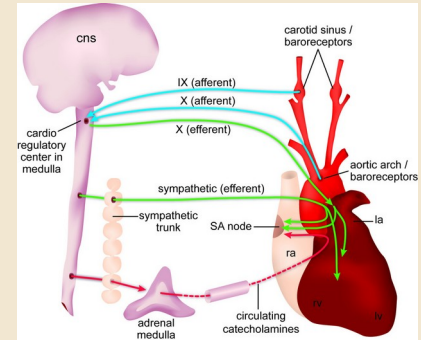
"Is my circulating volume / pressure at set point?"

Sensors:

baroreceptors (carotid, aortic, atrial)
juxtaglomerular apparatus

Responses:

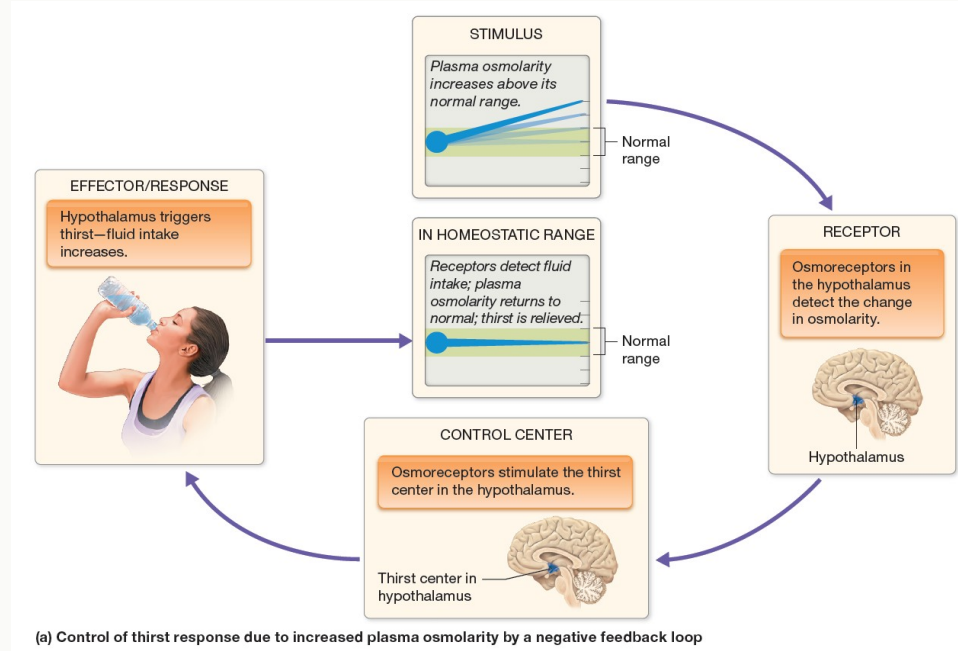
- RAAS / aldosterone
- Sympathetic
- ANP (high vol → brake)



Osmolarity Feedback — ADH + Thirst

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CONCEPT



Loop sequence:

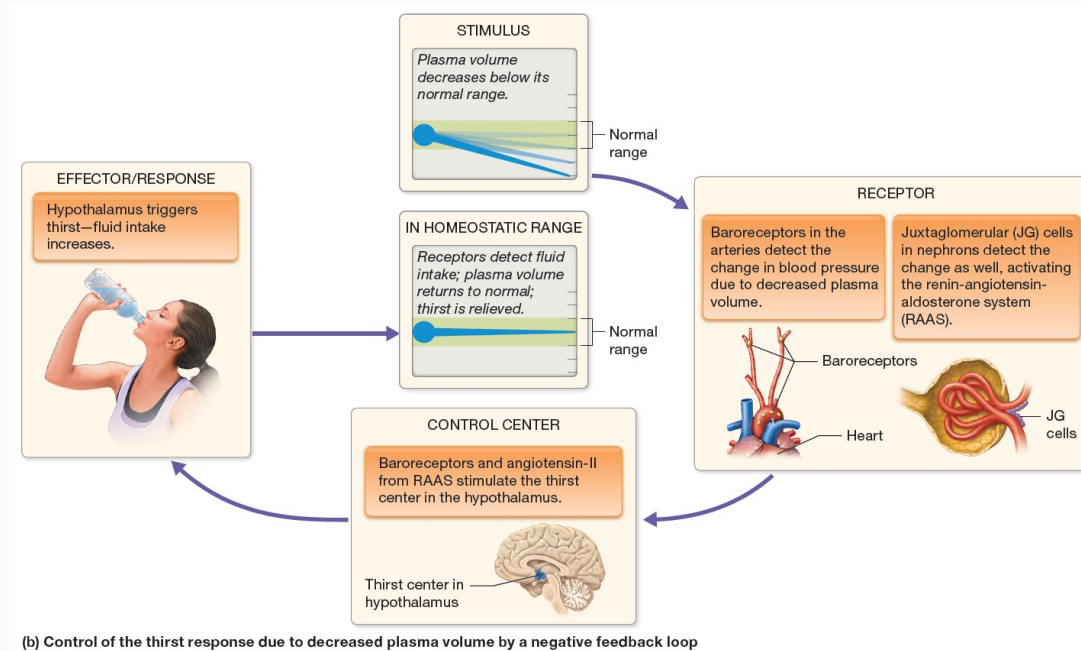
1. Plasma osm $\uparrow$
2. Osmoreceptors fire
3. Posterior pituitary ADH
4. AQP-2 in collecting duct
5. Free water reabsorbed
6. Plasma osm $\downarrow \rightarrow$ loop off

Amerman Fig. 25.7a

Volume Feedback — RAAS + Non-Osmotic ADH

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CONCEPT



Loop sequence:

1. Plasma volume ↓
2. Baroreceptors + JG cells
3. RAAS → angiotensin II
4. Aldosterone (Na⁺ retention)
5. Non-osmotic ADH
6. Volume restored

Amerman Fig. 25.7b

What's Happening at the Cellular Level?

STEP

Use last week's framework and what we just learned about water shifts.

- 1 Which microscopic structure(s) on the nephron are damaged in Reyna?
- 2 Sketch a glomerulus and identify the abnormality (compare to Santiago's HUS).
- 3 How does that damage explain albumin in her urine and her leg edema?
- 4 What's making her pee so much? (Hint: think osmotic diuresis and her diabetes.)

The Edema Cascade

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CONCEPT

- 1 Damaged glomerulus leaks albumin into urine (100 mg/24hr)
- 2 Plasma albumin slowly drops over months/years
- 3 Plasma oncotic pressure (π , like colloid osmotic pressure) drops
- 4 Capillary HP $>$ $\pi \rightarrow$ fluid leaks into interstitium
- 5 Pitting edema, especially in legs (gravity)

Form following function, in slow motion.

Why Reyna Pees So Much

CONCEPT

Reyna's symptom: "I find myself needing the restroom quite a lot."

- 1 Years of high blood glucose
- 2 Filtered glucose load > tubule's reabsorption capacity (SGLT saturated)
- 3 Glucose stays in tubule lumen → drags water osmotically (OSMOTIC DIURESIS)
- 4 Polyuria → ECF volume drops + osmolality climbs
- 5 ADH ↑ + thirst ↑ → Reyna also drinks more (POLYDIPSIA)

Polyuria + polydipsia in a diabetic = poorly controlled disease.

Reyna's Story Is Not Color-Blind

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HEALTH EQUITY

Who develops diabetic kidney disease?

- T2DM and diabetic nephropathy fall disproportionately on Black, Latinx, Native American, Pacific Islander, and South Asian patients in the U.S.
- Black Americans are 3–4× more likely to develop kidney failure from diabetes than white Americans

Why? Three layers

- Genetic — APOL1 variants (African ancestry) accelerate CKD progression
- Socioeconomic — food access, work conditions, healthcare cost
- Structural — delayed screening, fewer specialty referrals, weaker preventive continuity of care

The biology is universal. The clinical care is not — yet.

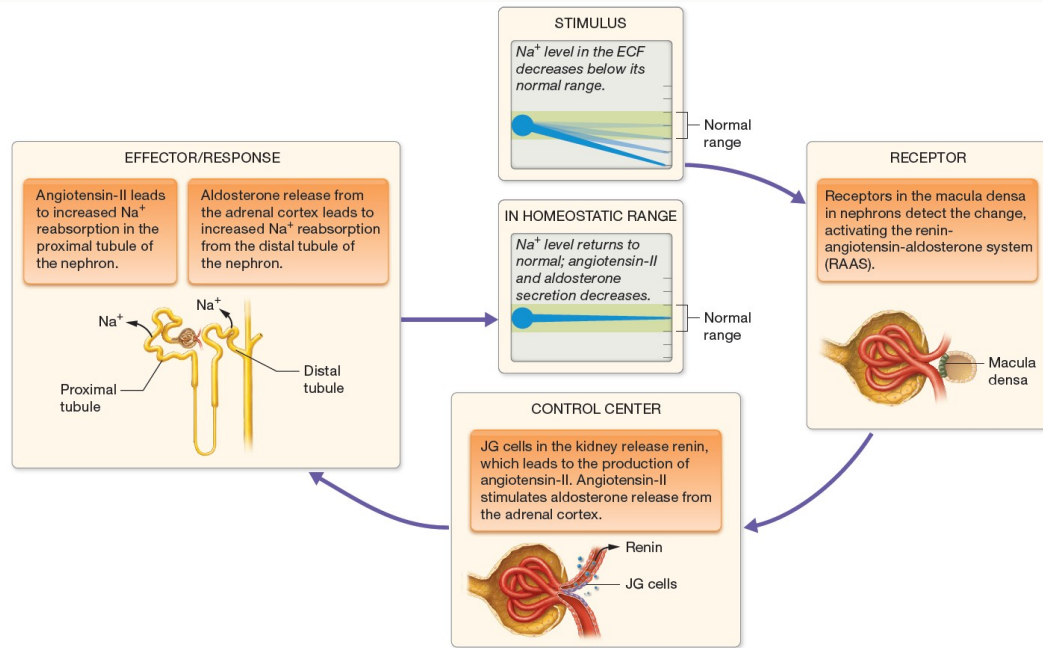
Block 4

Why Reyna's BP Is High + Step 4 (Treatment)

Why Reyna's Blood Pressure Is High

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CONCEPT



Damaged kidney →

- Reduced GFR
- Macula densa senses $\downarrow \text{Na}^+$ delivery
- Renin released
- Angiotensin II vasoconstricts
- Aldosterone holds Na^+ + water
- BP rises

Slow positive-feedback spiral

HTN damages glomeruli → more RAAS → more HTN

Amerman Fig. 25.8 — RAAS feedback for Na^+ balance

Propose a Treatment Plan

ACTIVITY

With your team:

- 1 What is the likely first-line treatment for Reyna's condition?
- 2 From a physiology perspective, how does it work? (Be specific about where in the nephron.)
- 3 What outcome would you expect over 6 months for: BP, urine albumin, edema, eGFR?

Hint: think back to Wk 3 AA-3b on ACE inhibitors — the same drug, with new framing.

ACE Inhibitors: Last Week vs. This Week

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A&P REVIEW

Last week (AA-3b)

ACE inhibitor blocks angiotensin II → efferent arteriole dilates → glomerular HP drops → ↓ GFR.

This sounds bad.

For Reyna — RENOPROTECTIVE

Lowering glomerular pressure protects damaged filters from further injury.

Small short-term GFR drop. Big long-term gain.

Gold standard for diabetic kidney disease + hypertension.

Same drug. Two framings. Same physiology.

Predict Reyna's 6-Month Trajectory

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ACTIVITY

Start her on an ACE inhibitor today. Predict each in 6 months:

Blood pressure	↓
Urine albumin	↓
Pitting edema	↓
eGFR	small initial ↓ , then stable

Slowing the disease IS the win — full reversal is not the goal here.

Knowing the Drug Isn't the Same as Getting It

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HEALTH EQUITY

ACE inhibitors are remarkable

- ~\$5/month generic
- Slow CKD progression by years
- Reduce all-cause mortality in proteinuric kidney disease
- Single most effective intervention for slowing diabetic nephropathy

But access is uneven

- Uninsured patients often go years without them
- Black and Latinx patients prescribed ACE inhibitors at lower rates even with equivalent disease severity
- Rural patients face fewer specialty referrals to start them

Knowing the science is part one. Advocating for access is part two.

Three Takeaways

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CONCEPT

1

ICF is bigger than ECF. Most fluid lives inside cells.

2

Water (osmolality) and sodium (volume) are regulated by two different sensor systems with different hormones.

3

Reyna's edema, polyuria, and high BP are all downstream of one slow process: damaged glomeruli over years.

Wednesday

Same framework — dial turned to acid-base

Meet Oscar — and learn the 30-second ABG read.